SOIL EROSION AND CONSERVATION · NOTE ST01

Permanent Conservation Structures — Purposes, Planning, Design Stages and Types of Spillway

Course topics: SWCE-503 T1, SWCE-503 T2, SWCE8.12 L8

Why permanent structures

- Permanent structures of earth, masonry or reinforced concrete are efficient supplements to soil and water conservation and gully control. Use them where vegetation cannot handle concentrated runoff, temporary structures cannot reclaim the gully, water must be stored (domestic use, irrigation, livestock) or life and property must be protected. Vegetation is also at the mercy of climate, diseases and pests.
- Well designed, built and maintained, they last long and are dependable. Their main functions: halt the advance of overfalls at gully heads; stabilise the grade so the gully becomes a regulated waterway; and (drop-inlet structures) control flow downstream.
- They are costly, so use them only when vegetative measures and simple field structures are inadequate — generally in medium to large gullies with medium to large drainage areas — planned on a watershed basis and built only after careful engineering investigation and detailed plans.

Purposes: control of gully advance, gully grades and stabilisation; water storage; sediment storage; surface-water inlets to main drains; detention to cut peak runoff (flood prevention); water-level control; stream-bank, tide and shore protection. The basic gully-stabilisation structures are **check dams, drop spillways, chute spillways and drop-inlet spillways**, each suited to particular sites.

Design requirements

- A gully-control structure must pass the design discharge, dissipate the kinetic energy of the flow within itself, and protect itself and the downstream channel. The **primary causes of failure** are insufficient hydraulic capacity and insufficient energy dissipation.
- Design follows investigation of the approaching runoff, the structure's function, the site and the downstream flow — there is no standard solution.

Data and investigations

- **Data:** topographic maps of the contributing watershed and the adjoining downstream area; soil and rainfall information; watershed characteristics (topography, vegetative cover, land use). Survey intensity depends on the structure's type, importance and purpose.
- **Preliminary investigations** — reconnaissance, collection and analysis of existing data, and an outline survey — compare sites and establish techno-economic feasibility. Reliable data mean less over-design and a cheaper structure.
- **Outline surveys:** temporary and permanent bench marks, channel grades, gully cross-sections at possible sites, regular and unusual sections, and the levels and positions of existing structures.
- **Sources:** National Bureau of Soil Survey and Land Use Planning (Nagpur); All India Soil Survey Organisation (New Delhi); state soil survey organisations; India Meteorological Department (Pune), with rainfall data from over 3000 rain-gauge stations.

Three stages of design

Stage	Content
Hydrologic design	Design runoff rates and flood volumes from rainfall (amount, intensity, frequency, seasonal distribution) and watershed characteristics, using meteorological records. Permanent structures carry the runoff of the heaviest rain expected once in 25–50 years or more, matched to the structure's life (life 25 years → 25-year flood). Drop-inlet spillways also need total runoff volume, inflow–outflow hydrographs and stage–storage curves for flood routing. Use the rational method for drop structures and chutes on watersheds up to about 1300 hectares, and the Curve Number method for larger catchments
Hydraulic design	Dimensions to handle the peak runoff, outflow design, storage capacity (drop-inlet structures), effects on the upstream and downstream channels, and dissipation of the energy released by the drop in water level
Structural design	Strength and stability against water pressure (static and dynamic), overflow forces and forces from flow beneath the structure (seepage, subsurface flow). The structure must stay stable, resist sliding by shear resistance and friction from its own weight, and have internal strength at least equal to the external stresses

Foundations: build on firm ground; avoid wet foundations (or drain them artificially); strip surface soil and organic matter; ensure a good bond between structure and foundation.

Spillway terms

Term	Meaning
Spillway	Passage that disposes of surplus water from the headwater pool or entrance channel — over the dam or upstream of it in a side ridge; an open channel or a closed conduit
Principal (main, mechanical, structural) spillway	Passes the design flood of the shorter recurrence interval
Auxiliary spillway	Discharges part of the larger, rarer flows; of structural material or vegetation, with its crest above the main spillway — works only when the main spillway's capacity is exceeded
Emergency spillway	Carries flows larger than the design storm and relieves the other spillways if they clog or are damaged; set at the design water level; usually a vegetated waterway
Stepped (cascade) spillway	Steps on its face dissipate energy
Overflow (overfall, straight-drop, ogee) spillway	Flow drops freely from the crest — for gravity and thin-arch overflow dams, with a nearly vertical downstream face; free-discharging, weir-controlled flow. The **ogee** has a reversed, elongated-S curve whose upper part follows the lower nappe of a ventilated sheet from a sharp-crested weir, so flow clings to the face when air is kept out from under it
Side-channel spillway	The control weir runs along one side, roughly parallel to the discharge channel; flow drops into a narrow trough, turns about a right angle and enters the main channel
Chute (open-channel, trough) spillway	Conveys flow from the upper reach or reservoir to the downstream level through an open channel along the dam, an abutment or a saddle — common on earth-fill dams; inlet channel, control structure, discharge channel, outlet structure and outlet channel, with a straight centre line and uniform width
Drop-inlet (shaft) spillway	Flow enters over a horizontal top, drops down a vertical or sloping shaft and runs through a nearly horizontal conduit or tunnel to the river below — overflow weir, vertical transition, closed discharge channel. With a funnel-shaped inlet it is a **morning-glory (glory-hole)** spillway

- **Uncontrolled crests** release water automatically once the level rises above the crest, needing no attendant or gates. Gates are needed only for very long crests or extra head — usually not in conservation structures.
- **Freeboard** is the height from the design water surface (or, with an emergency spillway, from its crest) to the top of the settled embankment. It guards against waves, extraordinary floods, long-term settlement, breaches by burrowing animals, roots, traffic on the crest and other unforeseen conditions. For farm ponds, erosion-control structures, drop-inlet spillways and small irrigation reservoirs it is **30–60 cm**, determined by Stephenson's formula (Richey, C. B. et al., 1961).

Sources: ICAR eCourse 'Soil and Water Conservation Engineering' — Dr A. Mishra, reviewed by Dr B. C. Mal (IIT Kharagpur); AgriMoon e-book. US Bureau of Reclamation, Design of Small Dams and Hydraulic Design of Stilling Basins and Energy Dissipators (EM 25). Further reading: A. M. Michael and T. P. Ojha, Principles of Agricultural Engineering, Vol. II (Jain Brothers).

Practice MCQs

Q1. The two primary causes of failure of gully-control structures are

(a) poor materials and cracks
(b) insufficient hydraulic capacity and insufficient energy dissipation
(c) seepage and rodents
(d) settlement and wind

Q2. Permanent structures are designed for runoff of recurrence

(a) 2–5 years
(b) 5–10 years
(c) 25–50 years or more
(d) 1000 years

Q3. Rational method is used for drop structures and chutes on watersheds up to about

(a) 50 ha
(b) 300 ha
(c) 1300 ha
(d) 10,000 ha

Q4. A funnel-shaped drop-inlet spillway is called a

(a) side-channel spillway
(b) ogee spillway
(c) morning-glory spillway
(d) siphon spillway

Q5. The emergency spillway is usually a

(a) concrete chute
(b) vegetated waterway
(c) pipe
(d) ogee weir

Q6. Freeboard for farm ponds and small erosion-control structures is

(a) 5–10 cm
(b) 30–60 cm
(c) 1–2 m
(d) 3–5 m

Q7. Basic gully-stabilisation structures do NOT include

(a) drop spillways
(b) chute spillways
(c) drop-inlet spillways
(d) contour furrows

Answers

Q1 — (b) insufficient hydraulic capacity and insufficient energy dissipation
Q2 — (c) 25–50 years or more
Q3 — (c) 1300 ha
Q4 — (c) morning-glory spillway
Q5 — (b) vegetated waterway
Q6 — (b) 30–60 cm
Q7 — (d) contour furrows